Six Iterated Integrals over a Parabolic Cylinder Solid

Stewart, *Calculus: Early Transcendentals*, §15.7, Exercise 31

Problem

Express the integral

$$\iiint_E f(x, y, z)\, dV$$

as an iterated integral in six different ways, where E is the solid bounded by the surfaces

$$y = x^2, \qquad z = 0, \qquad y + 2z = 4.$$

Part (a)

What the symbols mean

- E is the solid region of space. The integral adds up the values of f over all points of E.
- dV is the volume element. In Cartesian coordinates it is the product of the three differentials, for example $dV = dz\, dy\, dx$. The order of the differentials must match the order of the integral signs: the innermost sign goes with the first differential.
- f is a given function of three variables. In the answers below, f always means $f(x, y, z)$.
- An *iterated integral* is a chain of three ordinary integrals. The outer integral runs over a fixed interval. The middle integral runs over an interval that may depend on the outer variable. The inner integral runs between two surfaces, written as functions of the two variables that are already fixed.
- A *projection* of E on a coordinate plane is the set of points (u, v) in that plane such that the line through (u, v) parallel to the remaining axis meets E. The projection gives the limits of the two outer integrals. A projection is also called a *shadow*.

Description of the solid

The surface $y = x^2$ does not contain z. It is a parabolic cylinder: the parabola $y = x^2$ in the xy-plane, moved parallel to the z-axis. The plane $z = 0$ is the floor, and the plane $y + 2z = 4$ is a tilted roof. The roof slopes down as y grows, and it has height 2 at $y = 0$ and height 0 at $y = 4$.

The roof meets the floor where $y + 2z = 4$ and $z = 0$ at the same time, that is along the line $y = 4$, $z = 0$. The parabolic cylinder meets the floor along the parabola $y = x^2$, $z = 0$. The

cylinder meets the roof where $y = x^2$ and $y + 2z = 4$ hold at once, that is along the curve

$$y = x^2, \qquad z = \frac{4 - x^2}{2}, \qquad -2 \le x \le 2.$$

This curve rises from the two ends $(\pm 2, 4, 0)$ to the highest point $(0, 0, 2)$. At the ends the curve meets the floor line $y = 4$, $z = 0$, so the solid closes there. The solid is one piece, and it tapers to a single point at each end.

The solid lies above the parabola in the xy-plane and below the roof. Because $z \ge 0$ and $y + 2z \le 4$ together give $y \le 4$, the same solid is

$$\begin{aligned} E &= \{(x, y, z) : -2 \le x \le 2,\ x^2 \le y \le 4,\ 0 \le z \le (4 - y)/2\} \\ &= \{(x, y, z) : 0 \le y \le 4,\ -\sqrt{y} \le x \le \sqrt{y},\ 0 \le z \le (4 - y)/2\} \\ &= \{(x, y, z) : 0 \le z \le 2,\ 0 \le y \le 4 - 2z,\ -\sqrt{y} \le x \le \sqrt{y}\}. \end{aligned}$$

The three lines solve the three surface equations for the remaining variable. Figure 1 shows the solid with its three bounding pieces.

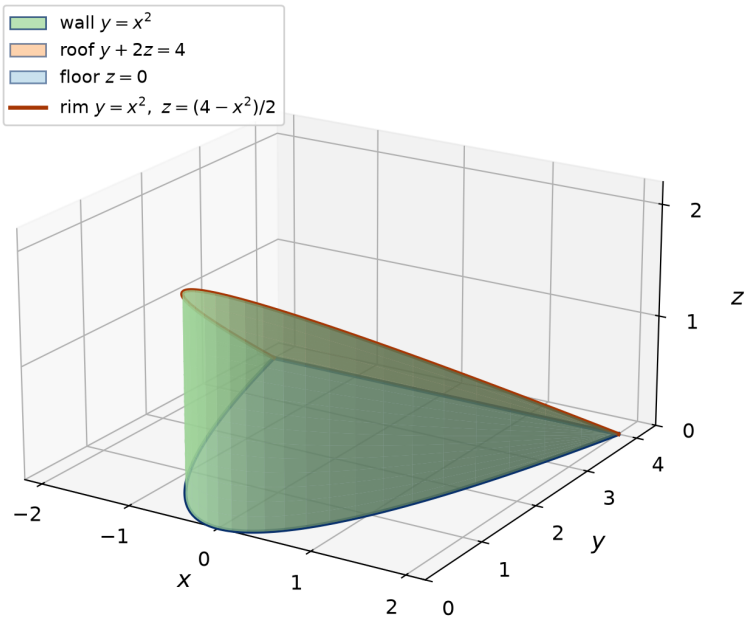


Figure 1: The solid E. The green curtain is the parabolic cylinder $y = x^2$. The orange sheet is the roof $y + 2z = 4$, and the blue sheet is the floor $z = 0$. The heavy dark blue curves are the parabola $y = x^2$ and the line $y = 4$ on the floor. The heavy red curve is the rim where the cylinder meets the roof, $y = x^2$ and $z = (4 - x^2)/2$. The rim runs from the point $(0, 0, 2)$ down to the two points $(\pm 2, 4, 0)$, where the solid closes.

The three projections

Each order of integration uses one coordinate plane twice. The two outer integrals run over a projection, and the inner integral runs along the line perpendicular to that projection. Three

planes are available, and each plane allows two orders, because the two outer integrals can be exchanged. Therefore six orders exist.

- *The xy-plane.* The projection is the part of the plane between the parabola $y = x^2$ and the line $y = 4$: the set $-2 \le x \le 2$, $x^2 \le y \le 4$. The same set is $0 \le y \le 4$, $-\sqrt{y} \le x \le \sqrt{y}$. The line perpendicular to this plane is vertical, so z is the inner variable. This gives the orders $dz\,dy\,dx$ and $dz\,dx\,dy$.
- *The yz-plane.* The projection is the triangle with corners $(y, z) = (0, 0)$, $(4, 0)$, $(0, 2)$: the set $0 \le y \le 4$, $0 \le z \le (4 - y)/2$. The line perpendicular to this plane is parallel to the x-axis, so x is the inner variable. This gives the orders $dx\,dz\,dy$ and $dx\,dy\,dz$.
- *The xz-plane.* The projection is the region below the parabola $z = (4 - x^2)/2$: the set $-2 \le x \le 2$, $0 \le z \le (4 - x^2)/2$. The line perpendicular to this plane is parallel to the y-axis, so y is the inner variable. This gives the orders $dy\,dz\,dx$ and $dy\,dx\,dz$.

Figure 2 draws the three projections and one slice of each.

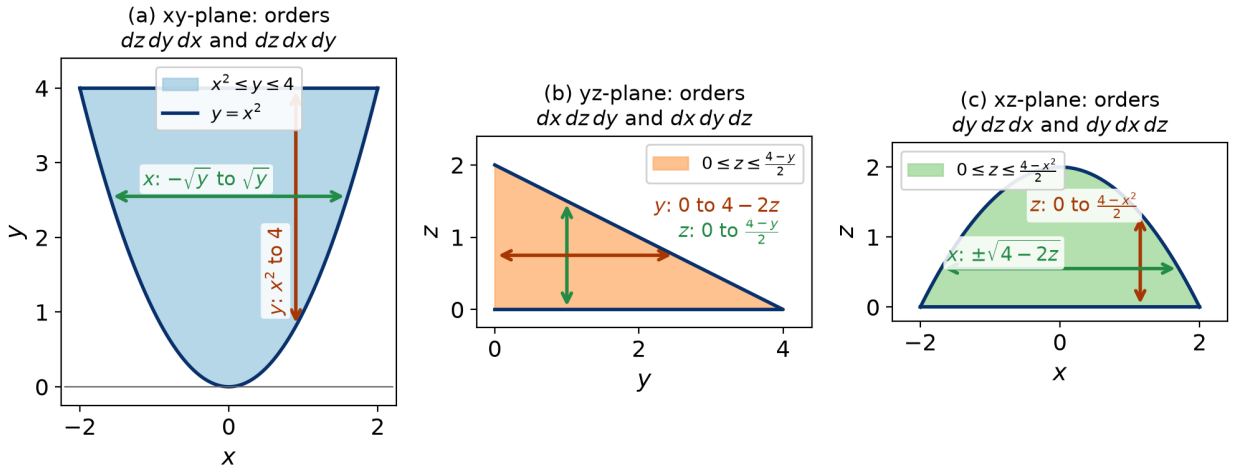


Figure 2: The three projections of E and the slices that give the limits. (a) The xy projection is the set $x^2 \le y \le 4$. The red arrow shows the range of y at a fixed x, which gives $dz\,dy\,dx$. The green arrow shows the range of x at a fixed y, which gives $dz\,dx\,dy$. (b) The yz projection is the triangle $0 \le y \le 4$, $0 \le z \le (4 - y)/2$. The green arrow shows the range of z at a fixed y, and the red arrow shows the range of y at a fixed z; both lines are parallel to the x-axis, so they give $dx\,dz\,dy$ and $dx\,dy\,dz$. (c) The xz projection is the set $0 \le z \le (4 - x^2)/2$. The red arrow shows the range of z at a fixed x, which gives $dy\,dz\,dx$; the green arrow shows the range of x at a fixed z, which gives $dy\,dx\,dz$.

The six iterated integrals

In every order the rule is the same. The inner limits come from the two surfaces that bound the solid along the inner axis. The outer two integrals run over the projection. A limit may use only variables that an outer integral has already fixed.

(1) **Order** $dz\,dy\,dx$**.** The outer variable is x. The solid meets the plane $z = 0$ at $y = x^2$, so $-2 \le x \le 2$. For fixed x the projection in the xy-plane runs from the parabola to the roof line, $x^2 \le y \le 4$. For fixed (x, y) the vertical line starts at the floor $z = 0$ and ends at the roof $z = (4 - y)/2$. That line meets E in one segment, so this order needs no split.

$$\iiint_E f\,dV = \int_{-2}^{2}\int_{x^2}^{4}\int_{0}^{(4-y)/2} f(x, y, z)\,dz\,dy\,dx.$$

(2) **Order** $dz\,dx\,dy$**.** The outer variable is y. Because $z \ge 0$ and $y + 2z \le 4$, the value of y lies between 0 and 4. For fixed y the parabola $y = x^2$ gives $x = \pm\sqrt{y}$, so $-\sqrt{y} \le x \le \sqrt{y}$. The roof height $(4-y)/2$ does not depend on x, so the inner limits are the same as in (1). The two outer integrals are exchanged, and both keep limits that do not depend on each other.

$$\iiint_E f\,dV = \int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}}\int_{0}^{(4-y)/2} f(x, y, z)\,dz\,dx\,dy.$$

(3) **Order** $dy\,dz\,dx$**.** The outer variable is x again. For fixed x the roof of the solid is the point where the cylinder meets the roof, at $z = (4-x^2)/2$. So $0 \le z \le (4-x^2)/2$. For fixed (x, z) the line parallel to the y-axis enters at the cylinder $y = x^2$ and leaves at the roof, where $y = 4-2z$. Both bounds are single graphs, so the inner limits are $x^2 \le y \le 4 - 2z$.

$$\iiint_E f\,dV = \int_{-2}^{2}\int_{0}^{(4-x^2)/2}\int_{x^2}^{4-2z} f(x, y, z)\,dy\,dz\,dx.$$

(4) **Order** $dy\,dx\,dz$**.** The outer variable is z, which runs from the floor $z = 0$ to the top of the rim $z = 2$. For fixed z the plane $y + 2z = 4$ cuts the solid in the region $x^2 \le y \le 4 - 2z$. That region is also $-\sqrt{4-2z} \le x \le \sqrt{4-2z}$ with $x^2 \le y \le 4 - 2z$. So x runs between those two square roots, and then y runs from the parabola to the plane.

$$\iiint_E f\,dV = \int_{0}^{2}\int_{-\sqrt{4-2z}}^{\sqrt{4-2z}}\int_{x^2}^{4-2z} f(x, y, z)\,dy\,dx\,dz.$$

(5) **Order** $dx\,dz\,dy$**.** Now the inner variable is x, so the two outer integrals run over the yz projection, the triangle $0 \le y \le 4$, $0 \le z \le (4 - y)/2$. For fixed (y, z) the line parallel to the x-axis starts on the cylinder at $x = -\sqrt{y}$ and ends on the same cylinder at $x = \sqrt{y}$.

$$\iiint_E f\,dV = \int_{0}^{4}\int_{0}^{(4-y)/2}\int_{-\sqrt{y}}^{\sqrt{y}} f(x, y, z)\,dx\,dz\,dy.$$

(6) **Order** $dx\,dy\,dz$**.** The outer variable is z again. At fixed z the plane $y+2z = 4$ gives $y = 4-2z$, so the yz projection is cut into the interval $0 \le y \le 4-2z$. The inner limits are the two halves of the cylinder, $-\sqrt{y} \le x \le \sqrt{y}$.

$$\iiint_E f\,dV = \int_{0}^{2}\int_{0}^{4-2z}\int_{-\sqrt{y}}^{\sqrt{y}} f(x, y, z)\,dx\,dy\,dz.$$

Order of integration	Inner limits	Middle limits	Outer limits
$dz\,dy\,dx$	$0 \le z \le \frac{4-y}{2}$	$x^2 \le y \le 4$	$-2 \le x \le 2$
$dz\,dx\,dy$	$0 \le z \le \frac{4-y}{2}$	$-\sqrt{y} \le x \le \sqrt{y}$	$0 \le y \le 4$
$dy\,dz\,dx$	$x^2 \le y \le 4-2z$	$0 \le z \le \frac{4-x^2}{2}$	$-2 \le x \le 2$
$dy\,dx\,dz$	$x^2 \le y \le 4-2z$	$-\sqrt{4-2z} \le x \le \sqrt{4-2z}$	$0 \le z \le 2$
$dx\,dz\,dy$	$-\sqrt{y} \le x \le \sqrt{y}$	$0 \le z \le \frac{4-y}{2}$	$0 \le y \le 4$
$dx\,dy\,dz$	$-\sqrt{y} \le x \le \sqrt{y}$	$0 \le y \le 4-2z$	$0 \le z \le 2$

Table 1: The limits of the six orders. Read the inner limits first. Each middle limit may use the outer variable, and each inner limit may use both outer variables.

One order evaluated in full

Check order (1) with $f = 1$. The inner integral is the height of the solid above the point (x, y):

$$\int_0^{(4-y)/2} dz = \frac{4-y}{2}.$$

The middle integral is the area of the section at fixed x. The section is a right triangle with legs $4 - x^2$ and $(4 - x^2)/2$, so

$$\int_{x^2}^4 \frac{4-y}{2}\,dy = \left[-\frac{(4-y)^2}{4}\right]_{y=x^2}^{y=4} = \frac{(4-x^2)^2}{4}.$$

The outer integral adds these areas:

$$V = \int_{-2}^2 \frac{(4-x^2)^2}{4}\,dx = \frac{1}{4}\int_{-2}^2 \left(16 - 8x^2 + x^4\right) dx = \frac{1}{4}\left(64 - \frac{128}{3} + \frac{64}{5}\right) = \frac{128}{15}.$$

The value $128/15 \approx 8.5333$ is the volume of E.

The same order gives the first moment in y. The middle integral is

$$\int_{x^2}^4 y\,\frac{4-y}{2}\,dy = \left[\frac{y^2}{2} - \frac{y^3}{6}\right]_{x^2}^4 = \frac{16}{3} - x^4 + \frac{x^6}{6},$$

and the outer integral is

$$\int_{-2}^2 \left(\frac{16}{3} - x^4 + \frac{x^6}{6}\right) dx = \frac{64}{3} - \frac{64}{5} + \frac{128}{21} = \frac{512}{35}.$$

Verification

Three scripts check the answer with three independent methods. The file `part1_step1.py` integrates all six orders exactly with `sympy`. The file `part1_step2.py` integrates the same six orders with a nested tanh-sinh quadrature rule, which uses 145 nodes in every layer, and then with a Monte Carlo sample of 6,000,000 points. The file `part1_step3.py` draws 400,000 random points and compares the geometric inequalities of E with the limit chain of every order.

The three test functions and their exact values are

$$\iiint_E 1\,dV = \frac{128}{15} \approx 8.533333333333, \qquad \iiint_E y\,dV = \frac{512}{35} \approx 14.628571428571,$$

$$\iiint_E \left(x^2+z^2\right) dV = \frac{512}{105} + \frac{4096}{945} = \frac{8704}{945} \approx 9.210582010582.$$

The second integral splits into $\int_E x^2\,dV = 512/105$ and $\int_E z^2\,dV = 4096/945$. The value of $\int_E x^2\,dV$ is $\int_{-2}^{2} x^2(4-x^2)^2/4\,dx$, because x^2 is constant on the section at fixed x. The value of $\int_E z^2\,dV$ is $\int_{-2}^{2}(4-x^2)^4/96\,dx$, because $\int_{x^2}^{4}\int_0^{(4-y)/2} z^2\,dz\,dy = (4-x^2)^4/96$.

Order of integration	$f=1$ (volume)	$f=y$	$f=x^2+z^2$
$dz\,dy\,dx$	8.533333333333	14.628571428571	9.210582010582
$dz\,dx\,dy$	8.533333333333	14.628571428571	9.210582010582
$dy\,dz\,dx$	8.533333333333	14.628571428571	9.210582010582
$dy\,dx\,dz$	8.533333333333	14.628571428571	9.210582010582
$dx\,dz\,dy$	8.533333333333	14.628571428571	9.210582010582
$dx\,dy\,dz$	8.533333333333	14.628571428571	9.210582010582
exact value	128/15	512/35	8704/945

Table 2: Values of the six iterated integrals for the three test functions. The numbers come from the tanh-sinh rule of `part1_step2.py`. The spread over the six orders is 1.8×10^{-15} for $f=1$, 3.6×10^{-15} for $f=y$ and 1.8×10^{-15} for $f=x^2+z^2$.

The Monte Carlo run gives $V = 8.533291 \pm 0.005777$, $\int_E y\,dV = 14.629799 \pm 0.011735$ and $\int_E (x^2+z^2)\,dV = 9.215237 \pm 0.008274$. Each interval contains the exact value. The random test of `part1_step3.py` finds no point where the limit chain of an order disagrees with the definition of E.

Answer

$$\boxed{\begin{aligned}
&(1)\quad \int_{-2}^{2}\int_{x^2}^{4}\int_{0}^{(4-y)/2} f(x,y,z)\,dz\,dy\,dx\\
&(2)\quad \int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}}\int_{0}^{(4-y)/2} f(x,y,z)\,dz\,dx\,dy\\
&(3)\quad \int_{-2}^{2}\int_{0}^{(4-x^2)/2}\int_{x^2}^{4-2z} f(x,y,z)\,dy\,dz\,dx\\
&(4)\quad \int_{0}^{2}\int_{-\sqrt{4-2z}}^{\sqrt{4-2z}}\int_{x^2}^{4-2z} f(x,y,z)\,dy\,dx\,dz\\
&(5)\quad \int_{0}^{4}\int_{0}^{(4-y)/2}\int_{-\sqrt{y}}^{\sqrt{y}} f(x,y,z)\,dx\,dz\,dy\\
&(6)\quad \int_{0}^{2}\int_{0}^{4-2z}\int_{-\sqrt{y}}^{\sqrt{y}} f(x,y,z)\,dx\,dy\,dz
\end{aligned}}$$

The volume of the solid is $\boxed{V = 128/15}$.

How to solve problems of this kind

1. Draw the solid first, and name every bounding surface. For this exercise the surfaces are a parabolic cylinder, a floor and a tilted roof.
2. Find the curve where two surfaces meet, because that curve is the edge of a projection. Here the cylinder meets the roof along $y = x^2$, $z = (4 - x^2)/2$, and this curve controls the xz projection.
3. Find the projection on the plane of the two outer variables. It gives the two outer limits, and a wrong projection is the most common error; Table 1 shows the three correct ones.
4. Choose a variable for the inner integral only after the projection is known. The inner limits are two surfaces solved for the inner variable. If a surface cannot be solved for that variable without a square root, check that the square root is real on the whole projection.
5. Test the inner limits with a line. Move a line parallel to the inner axis across the projection, and check that it meets the solid once. If the line meets the solid twice, split the solid into pieces.
6. Exchange the two outer integrals only when their limits do not depend on each other. A projection that is a triangle or a curved region usually needs a new description after the exchange, as in orders (2) and (5).
7. Check the answer with $f = 1$. The result must equal the volume, and the volume often has a second derivation, for example by slicing as in the text.
8. Check the answer with a nonconstant function. The sum $x^2 + z^2$ is a good choice, because each term follows from a section. Confirm the orders by exact and numeric methods.